

CLASS:BCA 3rd Sem

Notes PC Assembly & Troubleshooting

Unit-I

Brief history of computer on the bases Hardware, Computer system modules/ components and its operations, need of hardware and software for computer to work, **different hardware components within a computer** and **connected to a computer as peripheral devices**, different processors used for personal computers and notebook computers.

Unit-II

Perform installation, configuration, and upgrading of microcomputer/ computer: Hardware and software requirement, Assemble/setup microcomputer/ computer systems, accessory boards, types of motherboards, selection of right motherboard, Installation & replacement of motherboard, troubleshooting problems with memory.

Unit-III

Install/connect associated peripherals: Working of printers and scanners, Installation of printers and scanners, sharing a printer over a local area network, troubleshooting printer and scanner problems, troubleshooting hard drive problems. Drivers role and types

Unit-IV

Diagnose and troubleshooting of microcomputer/ computer systems hardware & software and other peripheral equipment: Approaches to solve a PC problem, troubleshooting a failed boot before the OS is loaded, different approaches to installing and supporting I/O device, managing faulty components.

Unit-1

HISTORY

Everyone says that computers are the latest buzzword that is happening in every walk of life. It is only the modern *avatara* of Computer; otherwise the history of computers goes back to the Vedic, Harappan, Greek, Egyptian and Japanese civilizations.

If we go by the simplest definition of computer then it is the instrument, which helps in computing. The idea of computing arisen due to the concept of counting. In the initial years of the development of human civilizations probably the fingers of our hands were the most common instrument to count the things, like one finger for one & two fingers for two & so on. Then many other things like sticks, small stones & other objects available naturally in abundant numbers were used to help counting.

Different civilizations grew & chosen their own methods of counting. As the trading started for various items & people came into contact with other civilizations, the need for counting & calculations grew many folds. Different civilizations invented different numerals to be used for counting. The invention of zero by India is very significant in this context. Different numerals used for counting have been shown in figure. With coming of numerals the system of writing them came into being. The decimal number system is now most prevalent universally. Slowly & slowly various civilizations contributed to the different aspects of mathematics. So, What we are doing today with the help of mathematics is the outcome of thousands of years of sustained research & growth.

The computers of today are also the outcome of thousands of years of brainstorming & application of science by different people around the world. The first instrument that is known in the history of computers was called Abacus.

Abacus: The literal meaning of Abacus is board or calculating table. Sometimes around 750 BC The Egyptians made this table of rectangular frame with rods & a divider as shown in Fig 2.1. The rods were having beads of different colors for counting. Hindus in India made the similar structure on sand & Greeks made it on the stones or slabs. The Chinese later perfected the Abacus.

There were total 10 columns. Moving the beads sideways-performed addition or subtraction. The place value of the digits of the numbers & position of beads helped in calculation process.

Fig 2.1 Abacus

Napier's bones: John Napier (1550-1617 A.D.) developed the idea of logs to transform multiplication problem to addition problem. The idea of logarithms later became the basis of the invention that is popularly known as the Slide rule. The Slide rule was invented in 1662. Napier also developed some set of rods for helping in multiplication. The reason of calling the invention as Napier Bones is that the rods were made of Bones.

Pascal's adding Machine: (1642 A.D.) A French mathematician Blaise Pascal invented a machine in 1642 for adding numbers. The characteristic of addition was the speed with which it 10

Gottfried Leibnitz's Calculator (1673 AD): A German mathematician, Gottfried Wilhelm Von Leibnitz (1645-1716 AD) improved Pascal's Adding machine. Now it was able to do the multiplication & division as well. The multiplications & divisions were performed through repeated additions & subtractions respectively.

Instead of wheels Leibnitz used stepped cylinder with nine teeth of varying lengths. It can also perform some other calculations like square roots etc.

Marie Jacquard's Loom with Punched Cards: A French weaver involved in designing clothes & making patterns or designs on clothes invented punched cards with holes to make beautiful designs. It was in 1801 AD when American Revolution was at its end. The presence or absence of a hole in the punched card represented two states for raising or lowering of the thread on the cloth. The whole process of weaving became automatic because punched card was storing information in the form of holes. This gave the idea of storing & retrieving information, which greatly influenced the future researchers in the field of computing.

Punched cards were in common use till 1990.

Babbage's Difference Engine (1822 AD): An English Professor of Mathematics Charles Babbage developed a Mechanical machine with the name Difference Engine. The machine was able to calculate various mathematical functions like logarithms, polynomials & algebraic. The precision in calculating these functions was quite good. It was giving the results up to 20 decimal places. 11 was done. So the machine was named as adding machine. For addition & subtraction it worked on clockwork mechanism principle. The adding machine consisted of numbered toothed wheels. Each tooth was having a unique position value. The rotation of wheels controlled the addition & subtraction operations. Carry-transfer was also done automatically in this.

Babbage's Analytical Engine (1833 AD): This was the machine that never became operational. But the idea & design of the machine given by Charles Babbage has some of the concepts being used in modern day computers. For this reason He is also called the Father of Computers. The meticulous & methodical design of the Analytical engine has the capability to perform comparisons. The features like Central processor, storage area, memory & input-output devices were incorporated in the design. It also has the capability to follow some machine sequences. The most fascinating of all was the permission to change the numbers & instructions already stored in the machine. The programs were to be stored in punched cards to be read by the machine. Lady Augusta Ada Lovelace helped Babbage in organizing the programs into the analytical machine concept. So she is also called as the First Programmer.

Herman Hollerith's Card Reader (1877 AD): An American Statistician Herman Hollerith, working in the department of Census fabricated the dream of Charles Babbage. He was annoyed at the time taken to calculate & organize the data of a census. It generally took around 10 years to calculate the data by which time the next census became due. By his efforts he developed the machine & used it for census data calculations. The data was now calculated in three years instead of 10 years. He used punched cards for input, output & instructions.

Hollerith founded a tabulating machine company that later merged with others to become the IBM (International Business Machines Corporation), one of the biggest companies in the field of computers.

Fig 2.5 Hollerith's Card Reader

Mark-I (1937 to 1943 AD): With the help of IBM Engineers Prof. Howard Aiken (1900-1973) constructed a computer named Mark-I. This could multiply numbers in seconds, which was a record at that time. It was first electromagnetic computer that could calculate according to pre-programmed instructions automatically without any need of human intervention. It was a huge machine and occupied several rooms. The inside of the computer had the electrical wires that were hundred miles long. Several electromechanical relays & mechanical counters were used for arithmetic calculations.

ENIAC: (1946 AD) this was the first electronic Computer developed by a team lead by Prof. Eckert and Mauchy at the University of Pennsylvania in USA. Electronic Numerical Integrator And Calculator called ENIAC used vacuum tubes as switching devices. It was having around 18,000 Vacuum tubes, 70000 registers, 10000 capacitors and 600 switches. It used 150,000 watts of electricity & cost \$ 400,000. It occupied around 5000 square feet of space. It took around 200 microseconds for addition & 2800 microseconds for multiplication. US army used it until 1955

to calculate the trajectories of missiles & other mathematical calculations. Instructions were given to the computer by External Plug Boards or switches. 13

EDVAC (1949 AD): By this time the binary number system was invented. So now it was simple to store in binary numbers instead of decimal numbers. Electronic Discrete variable automatic computer known as EDVAC also used the Von Neumann Architecture. This architecture advocates the concept of pre-stored instructions as programs. With this the operations became faster than before.

EDSAC (1949 AD): Prof. MV Wilkes working at Cambridge University built Electronic Delay And Storage Automatic Computer known as EDSAC. It used Mercury delay lines for storage. It also used binary number system & Von Neumann Architecture. This allowed easy implementation of loop structures in programs.

UNIVAC (1951 AD): Universal Automatic Computer known as UNIVAC was the first commercial computer developed by UNIVAC division of Remington Rand Company. It was used by General Electric besides other big companies. Initial applications of computers were in Science & Engineering where the calculations to be done are enormous.

IBM 650 (1955 AD): This computer made IBM as the leader in the field of computers. Thomas Watson, son of the IBM founder was responsible for introducing this computer with the help of IBM Engineer's.

IBM 700 Series (1955 AD): IBM 701, IBM 702 and IBM 703 were some of the computers from IBM 700 series. Each was having minor improvements from its predecessor but there was no major change in the design or technology being used.

After this many companies were involved in the manufacturing & development of computing machines. The computers gained the interest of the scientific community as a major tool for solving & performing various applications in different aspects of life. So there were thousands of models of different computers given by different companies in different countries. Instead of Discussing Each of them, they had been divided into five generations based on major changes in technology & design of computers.

First Generation (up to 1954): All the electronic computers described above are part of the first generation of computers. It started with ENIAC as the first electronic computer.

Major characteristics of the computers belonging to first generations were.

1. The computers used Vacuum Tubes.
2. The size of the machines was huge.
3. The speed of processing was very slow.
4. The space taken was too large.
5. The computers were very costly
6. Difficult to manufacture
7. Programming was very difficult
8. It was very difficult to remove the errors.
9. The computers were having all the components like Input, Output, CPU like the modern day computers
10. The use of these computers in Industrial or Commercial Sites was very less.
11. The scope of computers was thought to be in calculating the difficult calculations & doing the complex arithmetic.
12. Power consumption was more & the computers radiate a lot of heat.
13. Punched cards were used for Input & Output.
14. Only Machine level or Assembly language was used.

Second Generation (1955-1964)

IBM 1401(1958), IBM 1620(1960), IBM 7090(1960), IBM 7094(1961), IBM 7094I(1962), IBM 7094 II (1964), Control Data Corporation's CDC 1604(1961), CDC 3600(1964), Digital data corporation's PDP 1 (1957), PDP 5 (1963), PDP 8 (1965), UNIVAC 1108(1965) are all examples of the second generation computers.

1. These computers were using transistors for CPU components & ferrite cores for main memory.
2. As Transistors replaced the Vacuum tubes, so the size decreased as compared to First generation computers.
3. The performance of the computers increased because transistors consume one tenth of the power as compared to Vacuum tubes & generate less heat.
4. These used magnetic disks, Magnetic tapes & drums for secondary memory. Memory capacity was up to 100 Kilobytes (1 Kilo Byte= 1024 Bytes).
5. Floating point arithmetic was widely used in all these machines.
6. Use of High Level Languages Like FORTRAN, ALGOL & COBOL started in these computers.
7. The processing speed increased.
8. These were all batch processing systems.
9. The Computers were widely used in Science & Engineering Applications apart from many commercial applications.
10. New Professions like Programmers & system Analysts emerged during this period.
11. Academic programmes in computer science were started during this period.
12. Concept of operating system started evolving.

Third Generation Computers (1965-1974)

IBM/370 series (1970), CDC 7600 (1969), PDP 11(1970), ICL-2900 series (1968), Honeywell 6000 series, CYBER-175, STAR-100 are all examples of the third generation of computers. 15

The major characteristics of these are:

1. Instead of Transistors ICs (Integrated Circuits) were used for internal operations of CPU. Use of ICs eliminated wired interconnection between components. Initially these were using Small Scale Integrated Chips (SSI) that was having around 10 transistors per chip. In the later years computers were using Medium Scale Integrated Chips (MSI) that were having around 100 transistors per chip.
2. Semiconductor memories were used. So the memory capacity increased both at primary & secondary memory level. Memory capacity was now up to 100 Mega Bytes (1 Mega Byte = 1024 Kilo Bytes).
3. Concept of cache memory was also incorporated in third generation of computers.
4. There was further reduction in size & cost.
5. The reliability, efficiency & speed increased as compared to its predecessors.
6. Data Base Management Systems Emerged.
7. Business applications were now quickly moving towards automation.
8. Many Interactive Systems became feasible.

9. Mainframes were the highlights of this generation.
10. The CPUs were now able to carry 1 million instructions per second.
11. Multiprocessing, timesharing, multiprogramming systems were introduced.
12. Concept of networking evolved.

Fourth Generation Computers (1975-2002)

Motorola 68000 series, Intel Series, IBM 3090/600 (1988), IBM ES/9000(1996), VAX 8842(1988), HP 9000 Series, CRAY Y-MP (1988), IBM Deep Blue (1998), IBM PC/XT (1982), IBM PC/AT (1984), IBM PC/AAT (1986) are some of the examples of the Fourth generation computers.

The major characteristics of these are:

1. There were rapid improvements in the chip technology. In the initial years instead of MSI these computers used Large Scale Integrated chips that were having around 1000 transistors per second. In the late 80s these computers were using Very Large scale integrated chips that were having around 5000 transistors in a chip. The emergence of processor on a single chip has further increased the number of transistors on a single chip to around 500000.
2. The input & output devices like floppy disk, compact disk, video disk, touch screens, scanners, digital cameras, web cameras, highly improved printers became very common.
3. The size of the primary memory & secondary memory increased many fold. With improvements in the storage devices now hard disks are most popular media for storage. Now the size of the memory is up to 100 Giga Bytes (1 Giga Byte = 1024 Mega Bytes).
4. Separate processors are used for different processes like I/O processor, graphics processor etc.
5. Concept of distributed processing, parallel processing evolved.
6. Extremely powerful personal computers are now part of every home. Multimedia workstations emerged.
7. The size & cost of the Computers is continuously decreasing. The laptops & handheld computers are the examples of the miniaturization in computer of fourth generations.
8. Supercomputers are having the speed up to 1 billion instructions per second.
9. The way we do programming has also changed. In the initial years with the C language programming methodology changed to structured methodology. In the later years with C++ & Java now we are using object-oriented languages. Many interactive languages like Visual Basic, Visual C++, and Visual Age for Java also became popular.
10. Advanced databases incorporating Distributed databases concepts & having user-friendly graphical menus became prevalent.
11. With advancements in Communication technology & the coming of Internet computers changed the way of life. The way of doing businesses has changed. The concept of digital firm is evolving.
12. With the help of Fiber Optics networks the transmission speeds & quantity of data increased. It became possible to send Audio & Video as data.

Fifth Generation Computers

The Characteristics of these computers will be

1. Fifth generation computers will use Ultra Large Scale Integration Chips That will have millions of transistors on a single chip.
2. The speed of the doing computations will be around 1 trillion per second.
3. The memory size will go up to 100 Terra Bytes (1 Terra Byte = 1024 Giga Byte).

4. Many of the advanced countries are doing the projects related to this. Sample systems have already been prepared. These Computers will have Artificial Intelligence, so they may also be called as Expert Systems.
5. These Computers will have intelligence like human beings & will incorporate knowledge-based problem solving skills.
6. The input to these systems can be through speech & graphics. Similarly the output can be taken as speech.
7. These computers will be able to understand natural languages & will be able to convert one language from another language.
8. These computers will become more intelligent by time because they will have self-learning techniques.
9. Genetic Programming, Biotechnology, Biochips, Human like Robots will be the technologies that will evolve.
10. The computers will be part of mobile phones, handheld devices like watches etc.

Computer system modules/ components and its operations

What is a computer?

A computer is any machine that can be programmed to carry out a set of algorithms and arithmetic instructions.

Central Processing Unit, Input devices and Output devices. Input devices provide data input to processor, which processes data and generates useful information that's displayed to the user through output devices. This is stored in computer's memory.

Arithmetic Logic Unit (ALU)

Data entered into computer is sent to RAM, from where it is then sent to ALU, where rest of data processing takes place. All types of processing, such as comparisons, decision-making and processing of non-numeric information takes place here and once again data is moved to RAM.

Control Unit

As name indicates, this part of CPU extracts instructions, performs execution, maintains and directs operations of entire system.

Functions of Control Unit

Control unit performs following functions –

- ☐ It controls all activities of computer
- ☐ Supervises flow of data within CPU
- ☐ Directs flow of data within CPU
- ☐ Transfers data to Arithmetic and Logic Unit
- ☐ Transfers results to memory
- ☐ Fetches results from memory to output devices

Memory Unit

This is unit in which data and instructions given to computer as well as results given by computer are stored. Unit of memory is "Byte".

Input/output devices are required for users to communicate with the computer. In simple terms, input devices bring information INTO the computer and output devices bring information OUT of a computer system. These input/output devices are also known as peripherals.

Need of hardware software

Computer hardware is a collective term used to describe any of the physical components of an analog or digital computer. The term *hardware* distinguishes the tangible aspects of a computing device from software, which consists of written, machine-readable instructions or programs that tell physical components what to do and when to execute the instructions.

Hardware and software are complementary. A computing device can function efficiently and produce useful output only when both hardware and software work together appropriately.

Computer hardware can be categorized as being either internal or external components. Generally, internal hardware components are those necessary for the proper functioning of the computer, while external hardware components are attached to the computer to add or enhance functionality.

Computer Hardware Basics

Hardware – any physical device or equipment used in or with a computer system (anything you can see and touch).

External hardware

- External hardware devices (peripherals) – any hardware device that is located outside the computer.
- Input device – a piece of hardware device which is used to enter information to a computer for processing.
- Examples: keyboard, mouse, trackpad (or touchpad), touchscreen, joystick, microphone, light pen, webcam, speech input, etc.
- Output device – a piece of hardware device that receives information from a computer.
- Examples: monitor, printer, scanner, speaker, display screen (tablet, smartphone ...), projector, head phone, etc.

Internal hardware

- Internal hardware devices (or internal hardware components) – any piece of hardware device that is located inside the computer.
- Examples: CPU, hard disk drive, ROM, RAM, etc.

Computer Software Basics

1. Overview

Software not only makes your computer hardware perform important tasks, but can also help your business work more efficiently. The right software can even lead to new ways of working. It is therefore a crucial business asset and you should choose your software carefully so that it matches your business needs.

2. Business benefits of new software

Before investing in new software, you should think about what you want it to do for you. For example, you might want to:

- cut costs by automating routine tasks
- improve customer service
- enable your employees to work more efficiently
- communicate and collaborate electronically with suppliers or partners

Take time to talk to your employees, suppliers and customers who may have ideas for improving your business processes using IT.

Write down the objectives and potential benefits of new software, prioritising the list to work out the best returns on investment.

Choose software that will run on your current hardware, as long as this doesn't reduce potential benefits. You should include any hardware upgrade costs in your budgets.

Alternatively, look at outsourcing your software requirements - eg cloud computing - which could help reduce both your software and hardware costs.

or

Need of Software

Software, commonly known as programs or apps, consists of all the instructions that tell the hardware how to perform a task. These instructions come from a software developer in the form that will be accepted by the *platform* (operating system + CPU) that they are based on.

Practical computer systems divide software systems into two major classes:

□ **System software:** Helps run the computer hardware and computer system itself. System software includes operating systems, device drivers, diagnostic tools and more. System software is almost always pre-installed on your computer.

Types of System Software **1.System Management Software:** It is a collection of software that handles resources and provides general services for the other applications that run over them.

2.System Development Software: these are programs on which software programs rely to translate high-level language code to simpler machine-level code.for example complier .

□ **Application software:** Allows users to accomplish one or more tasks. It includes word processing, web browsing and almost any other task for which you might install software. (Some application software is pre-installed on most computer systems.) **1.Genral purpose Software:**

IT also known as end-user programs or productivity programs are software that helps the user in completing tasks such as doing online research, jotting down notes, setting an alarm, designing graphics, keeping an account log, doing calculations or even playing games. **2. Special purpose application software:** It is a type of software created to execute one specific task. For example, a camera application on your phone will only allow you to take and share pictures. Another example would be a chess game, it would only allow you to play chess or training software .

Computer software

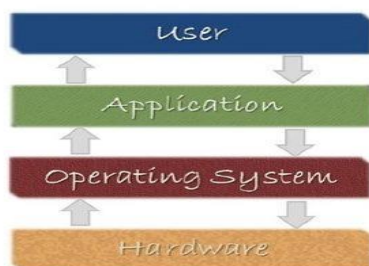
- Software – a set of instructions or programs that tells a computer what to do or how to perform a specific task (computer software runs on hardware).
- Main types of software – systems software and application software.

Application software

- Application software – a computer program that provides users with tools to accomplish a specific task.
- Examples of application software: word processing, spreadsheets, presentation, database management, Internet browsers, email programs, media players, accounting, pronunciation, translation, desktop publishing, enterprise, etc.

System Software

System software – it is designed to run a computer's hardware and application software, and make the computer system available for use. It serves as the interface between hardware, application software, and the user.



- Main functions of system software – allocating system resources, managing storage space, storing and retrieval of files, providing security, etc.
- Main types of systems software – operating system, device driver, utility software, programming software, etc.

Operating system (OS) – a software that controls and coordinates the computer hardware devices and runs other software and applications on a computer. It is the main part of system software and a computer will not function without it.

- Main functions of an operating system – booting the computer, managing system resources (CPU, memory, storage devices, printer, etc.), managing files, handling input and output, executing and providing services for application software, etc.
- Examples of operating system: Microsoft Windows, Apple iOS, Android OS, macOS, Linux, etc.

Device driver – a software program that is designed to control a particular hardware device that is attached to a computer.

- The main purpose of device driver – it acts as a translator between the hardware device and operating systems or applications that use it.
- It instructs computer on how to communicate with the device by translating the operating system's instructions into a language that a device can understand in order to perform the necessary task.
- Examples of device driver: printer driver, display driver, USB driver, sound card driver, motherboard driver, ROM driver, etc.

Utility software – a type of system software that helps set up, analyze, configure, strengthen, maintain a computer and performs a very specific task (e.g. antivirus software, backup software, memory tester, screen saver, etc.).

Different Computer External hardware

Input devices

Keyboard

Keyboard is the most common and very popular input device which helps to input data to the computer. The layout of the keyboard is like that of traditional typewriter, although there are some additional keys provided for performing additional functions.

Mouse

Mouse is the most popular **pointing device**. It is a very famous cursor-control device having a small palm size box with a round ball at its base, which senses the movement of the mouse and sends corresponding signals to the CPU when the mouse buttons are pressed.

Joystick

Joystick is also a pointing device, which is used to move the cursor position on a monitor screen. It is a stick having a spherical ball at its both lower and upper ends. The lower spherical ball moves in a socket. The joystick can be moved in all four directions.

Light Pen

Light pen is a pointing device similar to a pen. It is used to select a displayed menu item or to show objects on the monitor screen. It consists of a photocell and an optical system placed in a small tube.

Track Ball

Track ball is an input device that is mostly used in notebook or laptop computer, instead of a mouse. This is a ball which is half inserted and by moving fingers on the ball, the pointer can be moved.

Scanner

Scanner is an input device, which works more like a photocopy machine. It is used when some information is available on paper and it is to be transferred to the hard disk of the computer for further manipulation.

Microphone

Microphone is an input device to input sound that is then stored in a digital form.

The microphone is used for various applications such as adding sound to a multimedia presentation or for mixing music.

Bar Code Readers

Bar Code Reader is a device used for reading bar coded data (data in the form of light and dark lines). Bar coded data is generally used in labelling goods, numbering the books, etc. It may be a handheld scanner or may be embedded in a stationary scanner.

Output Devices

Monitors

Monitors, commonly called as **Visual Display Unit (VDU)**, are the main output device of a computer. It forms images from tiny dots, called pixels that are arranged in a rectangular form.

The sharpness of the image depends upon the number of pixels.

There are two kinds of viewing screen used for monitors.

- ☐ Cathode-Ray Tube (CRT)
- ☐ Flat-Panel Display

Cathode-Ray Tube (CRT) Monitor

The CRT display is made up of small picture elements called pixels. The smaller the pixels, the better the image clarity or resolution. It takes more than one illuminated pixel to form a whole character, such as the letter 'e' in the word help.

A finite number of characters can be displayed on a screen at once. The screen can be divided into a series of character boxes - fixed location on the screen where a standard character can be placed. Most screens are capable of displaying 80 characters of data horizontally and 25 lines vertically.

There are some disadvantages of CRT –

- ☐ Large in Size
- ☐ High power consumption

Flat-Panel Display Monitor

The flat-panel display refers to a class of video devices that have reduced volume, weight and power requirement in comparison to the CRT. You can hang them on walls or wear them on your wrists. Current uses of flat-panel displays include calculators, video games, monitors, laptop computer, and graphics display.

The flat-panel display is divided into two categories –

- ☐ **Emissive Displays** – Emissive displays are devices that convert electrical energy into light. For example, plasma panel and LED (Light-Emitting Diodes).

- ☐ **Non-Emissive Displays** – Non-emissive displays use optical effects to convert sunlight or light from some other source into graphics patterns. For example, LCD (Liquid-Crystal Device).

Printers

Printer is an output device, which is used to print information on paper.

There are two types of printers –

- ☐ Impact Printers

- ☐ Non-Impact Printers

Impact Printers

Impact printers print the characters by striking them on the ribbon, which is then pressed on the paper.

Characteristics of Impact Printers are the following –

- ☐ Very low consumable costs

- ☐ Very noisy

- ☐ Useful for bulk printing due to low cost

- ☐ There is physical contact with the paper to produce an image

These printers are of two types –

- ☐ Character printers

- ☐ Line printers

Character Printers

Character printers are the printers which print one character at a time.

These are further divided into two types:

- ☐ Dot Matrix Printer(DMP)

- ☐ Daisy Wheel

Dot Matrix Printer

In the market, one of the most popular printers is Dot Matrix Printer. These printers are popular because of their ease of printing and economical price. Each character printed is in the form of pattern of dots and head consists of a Matrix of Pins of size (5*7, 7*9, 9*7 or 9*9) which come out to form a character which is why it is called Dot Matrix Printer.

Advantages

- ☐ Inexpensive

- ☐ Widely Used

- ☐ Other language characters can be printed

Disadvantages

- ☐ Slow Speed

- ☐ Poor Quality

Daisy Wheel

Head is lying on a wheel and pins corresponding to characters are like petals of Daisy (flower) which is why it is called Daisy Wheel Printer. These printers are generally used for word-processing in offices that require a few letters to be sent here and there with very nice quality.

Advantages

- ☐ More reliable than DMP

- ☐ Better quality

- ☐ Fonts of character can be easily changed

Disadvantages

- ☐ Slower than DMP
- ☐ Noisy
- ☐ More expensive than DMP

Line Printers

Line printers are the printers which print one line at a time.

Laser Printers

These are non-impact page printers. They use laser lights to produce the dots needed to form the characters to be printed on a page.

Advantages

- ☐ Very high speed
- ☐ Very high quality output
- ☐ Good graphics quality
- ☐ Supports many fonts and different character size

Disadvantages

- ☐ Expensive
- ☐ Cannot be used to produce multiple copies of a document in a single printing

Different hardware components Connected to a computer as peripheral devices**Buses**

A bus is a subsystem that transfers data between computer components inside a computer or between computers. Each bus defines its set of connectors to physically plug devices, cards or cables together.

PCI

PCI (Peripheral Component Interconnect) is common in modern PCs. This kind of bus is being succeeded by PCI Express. Typical PCI cards used in PCs include: **network cards, sound cards, modems, extra ports** such as USB or serial, TV tuner cards and disk controllers.

Input

Input devices are absolutely crucial to computers. The most common input devices are mice and keyboards which barely every computer has. A new popular pointing device that may eventually replace the mouse is touch screen which you can get on some tablet notebooks. Other popular input devices include microphones, webcams, and fingerprint readers which can also be built in to modern laptops and desktops. A scanner is another popular input device that might be built-in to your printer.

Output

There are lots of different kinds of output devices that you can get for your computer. The absolute most common external output device is a monitor. Other very popular output devices are printers and speakers. There are lots of different kinds of printers and different sizes of speakers for your computer. Monitors are connected usually through the HD-15 connector on your video card. Printers are usually connected through a USB port. Speakers have their own audio out port built-in to the sound card.

USB

USB (Universal Serial Bus) is a serial bus standard to interface devices. USB was designed to allow many peripherals to be connected using a single

standardized interface socket and to improve the plug-and-play capabilities by allowing devices to be connected and disconnected without rebooting the computer.

Video Card

A video card (also known as graphics card) is an expansion card whose function is to generate and output images to a display. Some video cards offer added functions, such as video capture, TV tuner adapter, ability to connect multiple monitors, and others. Most video cards all share similar components.

Sound Card

A sound card is an expansion card that facilitates the input and output of audio signals to/from a computer under control of computer programs. Typical uses for sound cards include providing the audio component for multimedia applications such as music composition, editing video or audio, presentation/education, and entertainment.

Network Card

A network card is an expansion card that allows computers to communicate over a computer network. It allows users to connect to each other either by using cables or wirelessly. Although other network technologies exist, Ethernet has achieved near-ubiquity for a while now.

What is a Processor?

Definition: The processor is a chip or a logical circuit that responds and processes the basic instructions to drive a particular computer. The main functions of the processor are fetching, decoding, executing, and write back the operations of an instruction.

Types of Processors

There are different types of processors in the embedded system which include the following.

General Purpose Processor

There **are five types** of general-purpose processors they are, **Microcontroller, Microprocessor, Embedded Processor, DSP and Media Processor.**

1.Microprocessor

The general-purpose processors are represented by the microprocessor in embedded systems. There are different varieties of microprocessors available in the market from different companies. The microprocessor is also a general-purpose processor that consists of a **control unit, ALU, a bunch of registers, control registers and status registers.**

2.Microcontroller

The microcontroller is basically **a computer that comes in various packages and sizes.** The reading input and responding to output is the basic function of the microcontroller. Generally, it is known as General Purpose Input Output (GPIO). Some of the microcontrollers are Microchip Atmega328-AU, Microchip P1C16F877A-I/P, Microchip P1C16F1503-I/P, Microchip P1C16F671-I/SN, Microchip P1C18F45K22-I/P, etc.

3.Embedded Processor

An embedded processor is one type of processor which is **designed to control mechanical functions and electrical functions.** It consists of several blocks they are the processor, timer, an interrupt controller, program memory and data memory, power supply, reset and clock oscillator circuits, system application-specific circuits, ports and interfacing circuits.

4.Digital Signal Processor

The digital signal processor is one type of processor used for measuring, filtering and/or compress digital or analog signals. The signal processing means analysis and manipulation of signal. This processing can be done via computer.

Applications of DSP

The applications of the digital signal processor are

- ☐ Speech processing
- ☐ Image processing
- ☐ Medical processing
- ☐ Biometric Processing

5. Media Processor

The image/video processor is the media processor that is designed or created to deal with the data in real-time. The voice user interface and professional audio are the applications of the audio processor.

6. Core Processor

The core is the brain of the Central Processing Unit. There are different types of cores they are octa-core(8 processors, dual-core (2), quad-core(4) etc.

7. MultiProcessor

The multiprocessor is a computer with more than one CPU, each shares main memory, a computer bus, and peripherals to simultaneously process the programs and these systems are also known as tightly coupled systems. The advantages of multiprocessors are increased throughput, increased reliability and economy of scale. These processors are used when very high speed is required to process a large volume of data.

8. ASIC Processors

The application-specific integrated circuits are built for **specific applications**. These chips are small in size and consume low power. The design cost of ASIC is high and this is the main disadvantage. The application-specific integrated circuit chips are **used in satellites, modems, computers, etc.**